

Short answer question CO-1-AT-I

1. What is Data Science? Explain its primary objective!

→ Data Science is the field of extracting useful knowledge and insights from data using scientific methods, algorithms and statistics.

2. Why is Data Science important in today's digital world? Mention any four reasons.

- Helps in better decision-making
- Predicts future trends
- Improves business efficiency.

3. List any four benefits of Data Science in business and society

- Better customer understanding
- Increased business profits
- Improved healthcare service.

4. Four real-world application?

- Healthcare
- E-Commerce

• Transportation

5. Different facets of Data?

- Volume : A mount of data
- Velocity : Speed of data generations.
- Variety : Different types of Data

6. Structured (vs) Semi-Structured (vs) Unstructured

- Structured : Tables

- Semi-Structured : JSON, XML

- Unstructured : Images, Videos, emails

7. What is Big Data?

- Big Data is very large datasets; 5 Vs: Volume, Velocity, Veracity, Value and Variety.

8. Big Data ecosystem:

- A collection of tools for store, Process and analyze large datasets.

- hadoop

- Spark

- Hive

- H-Base

9. What are the major phases of the Data Science Process?

- Data collection

- Model Evaluation

- Data cleaning

- Deployment

- Data Analysis

10. What is data retrieval? Mention two common sources.

- Data retrieval is the Process of obtaining data from storage

- Databases

- APIs/website

11. why is data collection a stop in the Data Science Process?

- Accurate and relevant data leads to accurate analysis and better predictions.

12. what is data cleansing? List any four common data cleaning techniques.

- Data cleansing is the process of correcting or removing incorrect data.

→ Remove duplicate

→ handling missing values

→ Correct errors

→ removing outliers.

13. what is data integration? why is it necessary in Data Science?

- Data integrated combines data from multiple source into one dataset.

need: Improves data quality and provides a complete view.

14. Explain data transformation with suitable examples.

- Data transformation converts data into a suitable format for analysis.

Ex: Converting date "16/07/26" to "2026-07-06"

15. what is data analysis? Mention any four techniques used for data analysis?

- Data analysis is the process of examining data to find useful information.

→ Regression

→ classification

$$\frac{27}{30}$$

90%

→ Clustering

→ Data Visualization